

Safe Transfusion Practice: The Two Sample Rule of Group and Save Sampling

BACKGROUND

Blood transfusion errors can have catastrophic consequences. In 2010, JPAC advised **two-sample approach**.¹ This a critical safety measure to prevent ABO mismatch. This requires:

- Two separate venipunctures
- Independent patient identification and labelling

PROBLEM

Unsafe practices observed in the Emergency Department (ED):

- Collection of two samples from a **single venipuncture** (“double blood draw”)
- Incorrect labelling to simulate separate sampling events
- Increased risk of **ABO incompatible transfusion**

AIM

To eliminate the practice of double blood draw during ‘Group and Save’ sampling in the Emergency Department.

METHOD

Study Design: Model for Improvement using **Plan–Do–Study–Act (PDSA)** cycles

Duration: April 2024 – March 2025

Data Collection Strategy:

- Weekly 2-question survey of ED staff (~40 responses/week)
- Data collected at a fixed time point each week

1. In the last week, have you taken, or seen someone else take, two group and save samples from a single blood draw?
(If you don't see a 'thank you' pop-up with each press, please try tapping again)

Yes No

2. If yes, what was the staffing group of the person who did it?

Registrar Junior doctor Consultant Band 6, 7, 8 nurse Band 5 nurse Physicians assistant

Figure 1: Questionnaire used to collect information

Intervention Strategies:

i. PDSA Cycle 1

- Staff-wide educational email
- Engagement with Physician Associate (PA) lead

ii. PDSA Cycle 2:

- Re-engagement with PA leadership and ED matron
- Escalation to transfusion committee

DISCUSSION

- Educational and leadership engagement strategies were effective in reducing unsafe practices
- Sustainability is challenged by:
 - High staff turnover
 - Use of temporary and bank staff
 - Influence of practices from outside the ED
- System-level and cultural changes are required for long-term improvement

RECOMMENDATIONS

- Mandatory training on the two-sample rule for all ED staff
- Reinforcement during staff induction and ongoing education
- Regular audit and feedback cycles
- Expansion of interventions across other departments

RESULTS

Reduction in Unsafe Practices

- Both PDSA cycles resulted in a **statistically significant reduction** in ‘double blood draw’ events
- Complete elimination observed for 2 consecutive weeks** following PDSA Cycle 2
- A **rebound in unsafe practice** was observed 4 weeks after intervention

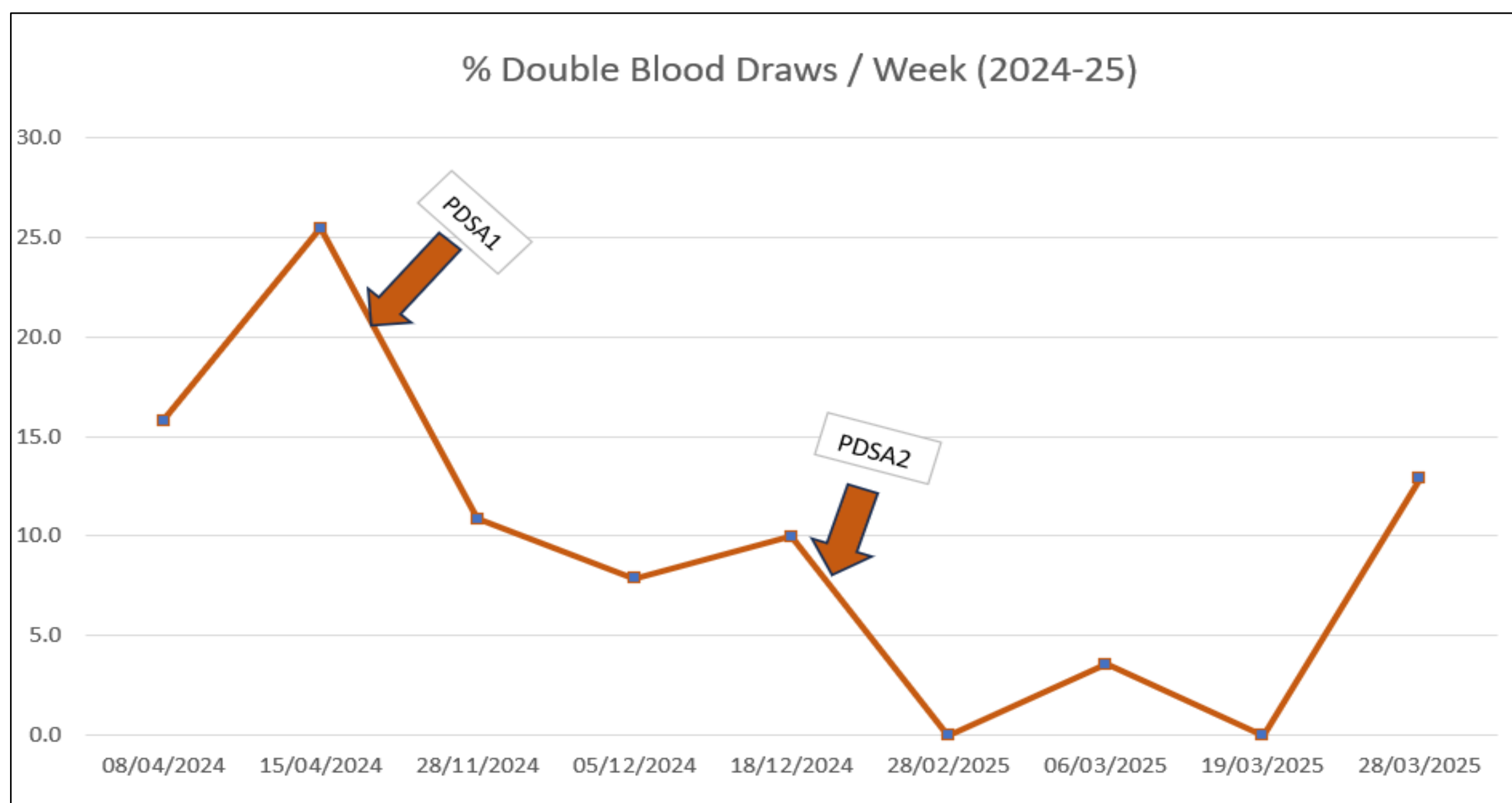


Figure 2: Weekly data collected during the project, where the percentage of ED staff who reported witnessing a Double Blood Draw event each week plotted against time. The 2 interventions are labelled as PDSA1 and PDSA2, with actions corresponding to that in Methods section

Identification of Staff Groups Involved

- Double blood draws were performed across **all staff categories**, including consultants
- Highest frequency observed among:
 - Physician Associates (PAs)
 - Band 5 nurses

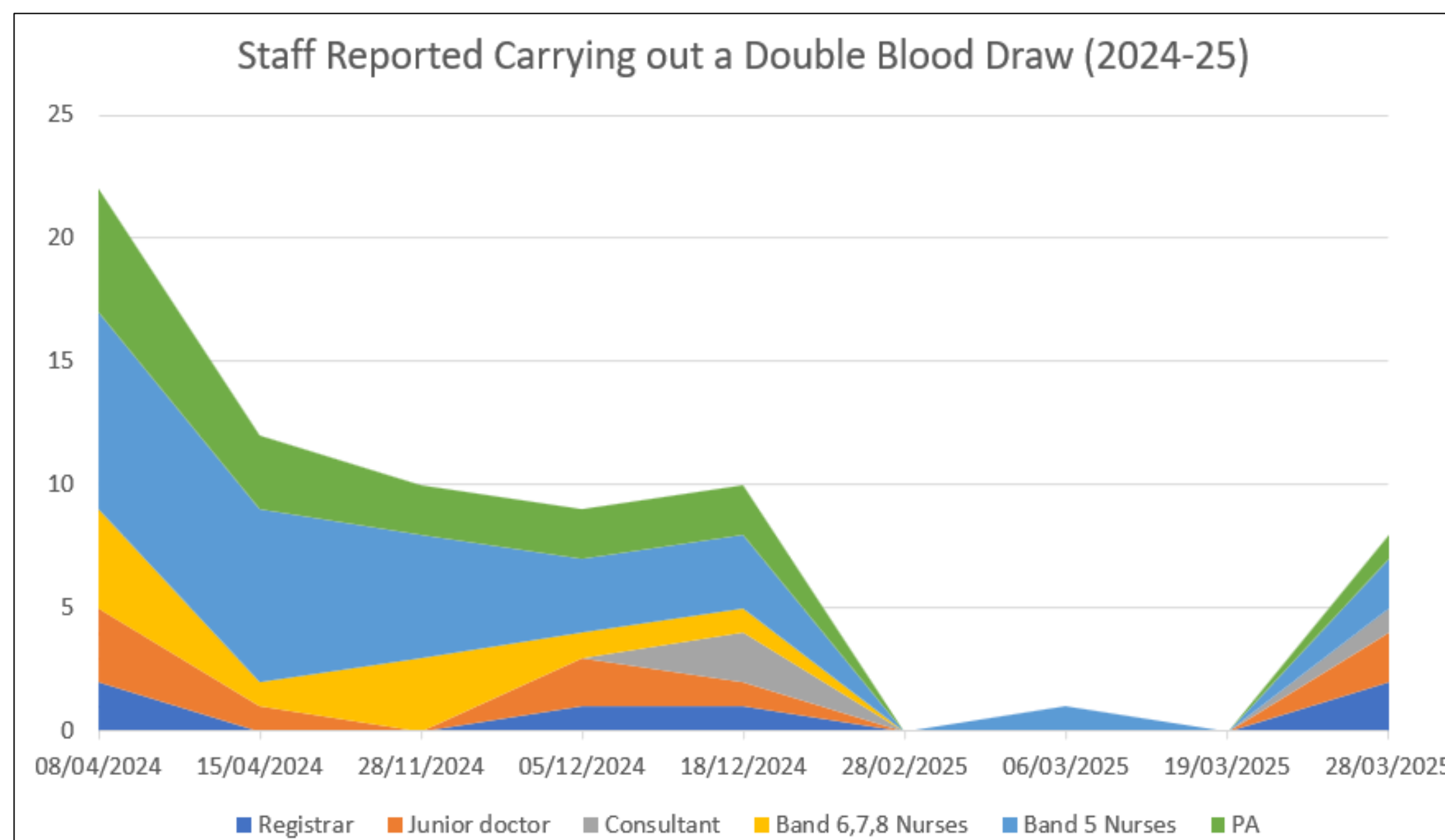


Figure 3: Category of staff performing Double Blood Draws, as reported by staff taking the questionnaire. Each coloured segments represent a staff category as labelled below the diagram. The X-axis height represents the absolute number of Double Blood Draw events that occurred during the week of data collection, plotted against time.

Staff Feedback

i. Physician Associates:

- Exposure to incorrect practices in other departments
- Pressure during emergency situations (e.g., trauma calls)

ii. Nursing Staff:

- Variable familiarity with protocol
- Practices influenced by previous workplace experience

KEY FINDINGS

- Targeted interventions can **rapidly improve compliance**
- Unsafe practices are **widespread across staff levels**
- Behavioral and system factors influence adherence
- Improvements may not be **sustained without reinforcement**

CONCLUSION

Targeted quality improvement interventions can significantly reduce unsafe transfusion-related practices. However, sustained improvement requires continuous education, reinforcement, and broader organizational engagement.

REFERENCES

- Principles of the EI Guidance Issued by the MHRA [online]. Joint United Kingdom (UK) Blood Transfusion and Tissue Transplantation Services Professional Advisory Committee (2010). Available from: <https://www.transfusionguidelines.org/regulations/clarification/it/principles-of-the-ei-guidance-issued-by-the-mhra>